

Calibration Behavior: 2024 vs. 2026

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Introduction

This report compares longitudinal calibration behavior between the 2024 and 2026 calibration datasets.

The analysis reproduces the drift analysis originally applied to the 2024 calibration data. For each anchor, calibration is compared with its calibration on the first day of the observation period.

Two measures of drift are examined:

1. **Instantaneous drift:** the absolute proportional difference between an anchor's current calibration and its calibration on Day 0.
2. **Maximum drift:** the largest instantaneous drift observed up to each point in time.

The analysis also examines the number of anchors that exceed 1%, 2%, 5%, 10%, and 20% deviation thresholds.

For the 2026 data, calendar dates are preserved throughout the analysis. Thus, the `days` variable represents actual elapsed calendar days rather than the number of calibration files retained after missing days were removed.

Overview

The report is organized as follows:

1. Load Data
2. Calculating Drift
3. Maximum Drift
 1. 90-99.9th Percentile Anchor Summary
 2. Geographic Context of IDs Associated with repeated ASNs in the 90-99.9 Range
 3. Visualizing Repeated ASN in 90-99.9th Percentile Range
 4. Flaw with Maximum Calibration Drift Visualization: Repeated ASN Behavior and Day 0 as Ground Truth
 5. Geographic Context of Single Anchor Associated ASNs in the 90-99.9 Range
 6. Filtered Maximum Drift (Day 0 Adjustment)
 7. Discussion on Adjusted Maximum Drift Visualization
4. Maximum Drift Threshold
 1. Observations/Discussion
 2. Anchor IDs that Exceeded 20%, 10%, and 5%
5. Instantaneous Drift
 1. 90-99.9th Percentile Anchor Summary
 2. Geographic Context of IDs Associated with repeated ASNs in the 90-99.9 Range
 3. Visualizing Repeated ASN in 90-99.9th Percentile Range
 4. Flaw with Maximum Calibration Drift Visualization: Repeated ASN Behavior and Day 0 as Ground Truth
 5. Geographic Context of Single Anchor Associated ASNs in the 90-99.9 Range
 6. Filtered Instantaneous Drift (Day 0 Adjustment: IDs from Instantaneous Drift 90-99.9th percentile)
 7. Discussion on Adjusted Instantaneous Drift Visualization
6. Instantaneous Drift Threshold
 1. Observations/Discussion
 2. Anchor IDs that Exceeded 20%, 10%, and 5%
7. Direct 2024 vs. 2026 Comparison
 1. Maximum Calibration Drift
 2. Instantaneous Calibration Drift
 3. Discussion
8. Notes for Interpretation/Process

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